

MARIINSK, Russian Federation

WMO: 295510

Lat: 56.183N

Lon: 87.683E

Elev: 473

StdP: 14.45

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.1	-29.0	-41.0	0.5	-35.8	-33.7	0.8	-28.7	25.5	15.5	21.1	15.0	0.9	60	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	84.6	67.0	81.3	65.4	78.3	64.3	70.1	79.5	68.1	77.3	66.2	75.4	4.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	101.2	74.9	64.8	93.3	72.8	62.8	87.0	70.8	34.5	79.6	32.8	77.4	31.3	75.7	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.2	18.2	15.7	DB	-40.9	90.1	7.1	2.5	-46.1	91.9	-50.3	93.3	-54.3	94.7	-59.5	96.5	
			WB	-41.0	73.9	6.6	3.2	-45.8	76.2	-49.7	78.1	-53.4	79.9	-58.2	82.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.9	0.9	7.4	21.8	37.6	50.5	62.1	65.7	60.5	48.9	36.4	18.4	6.5
	DBStd	25.04	16.65	14.08	11.57	9.80	9.69	7.52	5.66	6.66	8.12	9.74	15.06	16.08
	HDD50	6944	1523	1192	873	384	118	7	0	5	118	430	947	1347
	HDD65	11166	1988	1612	1338	822	459	138	60	169	485	886	1397	1812
	CDD50	1427	0	0	0	12	132	370	487	332	85	9	0	0
	CDD65	173	0	0	0	0	8	50	82	31	2	0	0	0
	CDH74	1808	0	0	0	0	152	578	739	298	41	0	0	0
	CDH80	410	0	0	0	0	37	145	162	58	8	0	0	0
Wind	WSAvg	5.8	5.7	6.1	6.6	6.9	6.7	4.9	3.8	4.0	5.1	6.3	6.7	6.7
Precipitation	PrecAvg	17.4	0.7	0.5	0.6	1.1	1.7	2.4	2.4	2.5	1.7	1.5	1.3	1.0
	PrecMax	26.2	1.8	1.3	1.3	2.5	3.2	5.4	8.0	5.6	4.1	3.0	3.1	2.6
	PrecMin	11.5	0.1	0.1	0.1	0.1	0.5	0.6	0.4	0.4	0.5	0.5	0.4	0.2
	PrecStd	3.4	0.4	0.3	0.3	0.6	0.7	1.3	1.7	1.1	1.0	0.7	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.4	36.7	49.8	69.2	85.2	88.6	88.0	86.1	79.8	64.9	48.4	36.8
		MCWB	32.4	32.6	41.9	51.9	62.0	65.8	70.1	69.0	61.5	50.9	42.9	33.4
	2%	DB	30.5	32.8	43.7	63.6	78.3	84.2	84.4	80.8	72.0	59.2	43.8	33.1
		MCWB	28.0	29.7	36.9	49.4	57.7	65.7	68.8	66.8	58.5	49.0	39.7	30.5
	5%	DB	27.2	30.1	40.2	58.6	73.4	80.5	81.1	77.2	67.1	54.6	39.4	30.2
		MCWB	25.2	27.6	34.6	46.9	55.6	63.7	67.0	64.6	56.1	46.8	36.4	28.3
	10%	DB	23.5	26.8	37.4	53.5	68.2	76.9	78.1	73.5	62.4	49.9	35.9	26.8
		MCWB	21.6	24.7	32.9	44.0	53.9	62.8	66.0	62.9	53.8	44.7	33.2	25.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.6	33.4	41.8	54.2	64.0	71.0	74.2	72.4	62.8	53.4	44.6	34.1
		MCDB	35.0	36.1	48.8	63.5	80.2	80.8	82.2	81.2	74.4	60.4	48.1	36.4
	2%	WB	28.4	30.2	38.1	50.9	60.9	68.2	71.0	68.4	59.6	50.2	40.0	31.0
		MCDB	30.4	32.5	42.3	62.3	74.3	78.3	79.8	77.2	69.8	57.8	42.7	32.6
	5%	WB	25.2	27.6	35.3	47.7	57.7	66.2	69.1	66.0	57.3	47.4	36.7	28.3
		MCDB	27.2	29.9	39.3	57.2	68.8	77.1	78.2	74.5	65.0	53.7	39.3	30.1
	10%	WB	21.8	24.8	32.9	44.4	55.0	64.4	67.3	64.0	54.9	44.7	33.3	25.0
		MCDB	23.7	26.8	37.0	53.5	66.5	74.5	76.0	71.9	61.2	49.7	35.5	26.9
Mean Daily Temperature Range	5% DB	MDBR	14.4	16.3	17.2	17.5	20.1	19.6	19.0	18.6	17.4	13.0	11.7	13.4
		MCDBR	13.1	13.5	18.4	25.9	28.9	26.8	24.3	24.8	26.7	21.5	12.5	14.1
		MCWBR	11.8	11.8	13.4	14.8	13.2	11.6	10.7	12.0	14.0	13.0	9.8	12.9
	5% WB	MCDBR	12.9	13.0	17.1	24.9	25.3	22.5	20.7	21.6	22.5	19.5	11.8	14.5
		MCWBR	11.8	11.7	13.3	15.5	13.1	11.5	11.0	11.5	12.7	12.8	9.5	13.6
Clear-Sky Solar Irradiance	taub	0.291	0.325	0.374	0.404	0.412	0.408	0.436	0.411	0.373	0.355	0.323	0.293	
	taud	1.994	1.968	1.904	2.055	2.190	2.292	2.213	2.262	2.314	2.215	2.032	1.945	
	Ebn at Noon	177	220	239	255	262	265	253	252	245	212	161	138	
	Edh at Noon	21	33	45	45	42	39	41	37	30	26	21	17	
All-Sky Solar Radiation	RadAvg	251	564	995	1382	1558	1848	1677	1346	878	442	215	149	
	RadStd	28	43	83	100	174	174	111	105	115	41	35	24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page